

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester V)
ENTERPRISE JAVA

Practical – III

Roll No.: T018	Name: Sachin Ramesh Mehta
Class: TY.IT	Batch: 1
Date of Assignment: 08/07/2026	Date/Time of Submission: 08/07/2026

AIM(A): Create a Servlet application to upload and download a file.

HTML

```
<!DOCTYPE html>
<html>
<head>
  <title>File Upload</title>
</head>
<body>
  <h1>File Upload Application</h1>
  <form action="FileUploadServlet" method="post" enctype="multipart/form-data">
    Select file to upload:
    <input type="file" name="file" required>
    <br><br>
    <input type="submit" value="Upload File">
  </form>
  <h2>Download Links</h2>
  <!-- Link to download files, ensure the path is correct -->
  <a href="FileDownloadServlet?filename=EJ_Practicals (2).pdf">
    Download the uploaded file
  </a>
  <br><br><br><br>
</body>
</html>
```

SERVELT: FileUpload

```
import java.io.File;
import java.io.FileOutputStream;
import java.io.IOException;
import java.io.InputStream;
import java.io.OutputStream;

import javax.servlet.ServletException;
import javax.servlet.annotation.MultipartConfig;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
```

```

import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.Part;
@WebServlet("/FileUploadServlet")
@MultipartConfig
public class FileUploadServlet extends HttpServlet {
    private static final String UPLOAD_DIR = "uploads";
    @Override
    protected void doPost(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        String uploadPath = getServletContext().getRealPath("")
            + File.separator + UPLOAD_DIR;
        File uploadDir = new File(uploadPath);
        if (!uploadDir.exists()) {
            uploadDir.mkdir();
        }
        Part filePart = request.getPart("file");
        String fileName = filePart.getSubmittedFileName();
        File file = new File(uploadPath + File.separator + fileName);
        try (InputStream inputStream = filePart.getInputStream();
            OutputStream outputStream = new FileOutputStream(file)) {
            byte[] buffer = new byte[4096];
            int bytesRead;
            while ((bytesRead = inputStream.read(buffer)) != -1) {
                outputStream.write(buffer, 0, bytesRead);
            }
        }
        response.getWriter().println("File uploaded successfully.");
    }
}

```

SERVELET:FileDownload

```

import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;
import java.io.OutputStream;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/FileDownloadServlet")
public class FileDownloadServlet extends HttpServlet {
    private static final String UPLOAD_DIR = "uploads";
    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        String fileName = request.getParameter("filename");
    }
}

```

```

        if (fileName == null) {
            response.getWriter().println("File name is missing.");
            return;
        }
        String filePath = getServletContext().getRealPath("")
            + File.separator + UPLOAD_DIR
            + File.separator + fileName;
        File file = new File(filePath);
        if (!file.exists()) {
            response.getWriter().println("File not found.");
            return;
        }
        response.setContentType("application/octet-stream");
        response.setHeader(
            "Content-Disposition",
            "attachment; filename=\"" + fileName + "\""
        );
        try (FileInputStream inputStream = new FileInputStream(file);
            OutputStream outputStream = response.getOutputStream()) {
            byte[] buffer = new byte[4096];
            int bytesRead;
            while ((bytesRead = inputStream.read(buffer)) != -1) {
                outputStream.write(buffer, 0, bytesRead);
            }
        }
    }
}

```

SERVELET:

```

import java.io.File;
import java.io.FileInputStream;
import java.io.IOException;
import java.io.OutputStream;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet("/FileDownloadServlet")
public class FileDownloadServlet extends HttpServlet {
    private static final String UPLOAD_DIR = "uploads";
    @Override
    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        String fileName = request.getParameter("filename");
        if (fileName == null) {
            response.getWriter().println("File name is missing.");
            return;
        }
    }
}

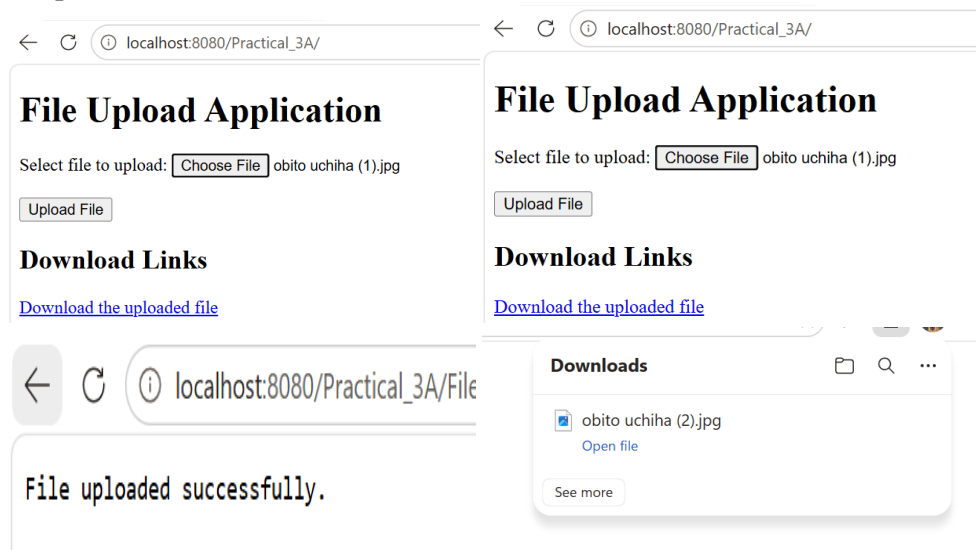
```

```

    }
    String filePath = getServletContext().getRealPath("")
        + File.separator + UPLOAD_DIR
        + File.separator + fileName;
    File file = new File(filePath);
    if (!file.exists()) {
        response.getWriter().println("File not found.");
        return;
    }
    response.setContentType("application/octet-stream");
    response.setHeader(
        "Content-Disposition",
        "attachment; filename=\"" + fileName + "\""
    );
    try (FileInputStream inputStream = new FileInputStream(file);
        OutputStream outputStream = response.getOutputStream()) {
        byte[] buffer = new byte[4096];
        int bytesRead;
        while ((bytesRead = inputStream.read(buffer)) != -1) {
            outputStream.write(buffer, 0, bytesRead);
        }
    }
}
}
}

```

Output:



AIM(B): Develop Simple Servlet Question Answer Application using Database.

Code:

```

<!DOCTYPE html>
<html>
<head>

```

```

<title>Java Quiz</title>
</head>
<body bgcolor="lightyellow">
  <center>
    <h1>Java Quiz Application</h1>
    <h3>Click below to start the quiz</h3>
    <a href="QuizServlet">START QUIZ</a>
  </center>
</body>
</html>

```

SERVLET:QuizServlet

```

package mypack;

import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/QuizServlet")
public class QuizServlet extends HttpServlet {
    public void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html>");
        out.println("<body bgcolor='lightblue'>");
        out.println("<center>");
        out.println("<h1>Java Quiz</h1>");
        out.println("</center>");
        out.println("<form action='ShowResult'>");
        try {
            Class.forName("com.mysql.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/qadb",
                "root",
                "root");
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("select * from quiz");
            int i = 1;
            while (rs.next()) {
                out.println("<hr>");
                out.println("<h3>Q" + i + ". "
                    + rs.getString("question")
                    + "</h3>");
                out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op1") + "'>" + rs.getString("op1")
                    + "<br>");
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}

```

```

        out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op2") + "'>" + rs.getString("op2")
+ "<br>");
        out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op3") + "'>" + rs.getString("op3")
+ "<br>");
        out.println("<input type='radio' name='" + i + "' value='" + rs.getString("op4") + "'>" + rs.getString("op4")
+ "<br>");
        i++;
    }
    con.close();
} catch (Exception e) {
    out.println(e);
}
out.println("<br>");
out.println("<input type='submit' value='Submit Quiz'>");
out.println("<input type='reset' value='Reset'>");
out.println("</form>");
out.println("</body>");
out.println("</html>");
}
}

```

SERVLET:ResultServlet

```

package mypack;

import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/ShowResult")
public class ShowResult extends HttpServlet {
    public void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        out.println("<html>");
        out.println("<body bgcolor='lightgreen'>");
        out.println("<center>");
        out.println("<h1>Quiz Result</h1>");
        out.println("</center>");
        try {
            Class.forName("com.mysql.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/qadb",
                "root",
                "root");
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("select ans from quiz");

```

```

int score = 0;
int i = 1;
while (rs.next()) {
    String correct = rs.getString("ans");
    String user = request.getParameter(""" + i);
    if (correct.equals(user)) {
        score++;
        out.println("<h3>Q" + i + " : Correct</h3>");
    } else {
        out.println("<h3>Q" + i + " : Incorrect</h3>");
    }
    i++;
}
out.println("<hr>");
out.println("<h2>Your Score : " + score + " / " + (i - 1) + "</h2>");
con.close();
} catch (Exception e) {
    out.println(e);
}
out.println("<br><a href='index.html'>Take Quiz Again</a>");
out.println("</body>");
out.println("</html>");
}
}

```

prac_3b.XML

```

<?xml version="1.0" encoding="UTF-8"?>
<web-app xmlns="http://xmlns.jcp.org/xml/ns/javaee"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://xmlns.jcp.org/xml/ns/javaee
        http://xmlns.jcp.org/xml/ns/javaee/web-app_3_1.xsd"
    version="3.1">
    <servlet>
        <servlet-name>QuizServlet</servlet-name>
        <servlet-class>mypack.QuizServlet</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>QuizServlet</servlet-name>
        <url-pattern>/QuizServlet</url-pattern>
    </servlet-mapping>
    <servlet>
        <servlet-name>ShowResult</servlet-name>
        <servlet-class>mypack.ShowResult</servlet-class>
    </servlet>
    <servlet-mapping>
        <servlet-name>ShowResult</servlet-name>
        <url-pattern>/ShowResult</url-pattern>
    </servlet-mapping>
</web-app>

```

Output:

Java Quiz Application

Click below to start the quiz

[START QUIZ](#)

Java Quiz
Q1. Which keyword is used to inherit a class in Java? ☐ implements ☒ extends ☐ inherit ☐ using
Q2. Which method is the entry point of a Java program? ☐ start() ☐ main() ☒ run() ☐ init()
Q3. Which package contains Scanner class? ☐ java.io ☐ java.util ☒ java.lang ☐ java.sql
Q4. Which is not a primitive data type? ☐ int ☐ float ☒ String ☐ char
Q5. Which exception occurs when a number is divided by zero? ☐ IOException ☐ NullPointerException ☒ ArithmeticException ☐ ClassNotFoundException ☐ FileNotFoundException

Quiz Result

Q1 : Correct

Q2 : Correct

Q3 : Correct

Q4 : Correct

Q5 : Incorrect

Your Score : 4 / 5

[Take Quiz Again](#)

```
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 1
Server version: 5.1.40-community MySQL Community Server (GPL)

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE qadb;
Query OK, 1 row affected (0.33 sec)

mysql> USE qadb;
Database changed
mysql> CREATE TABLE quiz
-> (
-> qno varchar(5) PRIMARY KEY,
-> question varchar(200),
-> op1 varchar(100),
-> op2 varchar(100),
-> op3 varchar(100),
-> op4 varchar(100),
-> ans varchar(100)
-> );
Query OK, 0 rows affected (0.31 sec)
```



```

C:\Program Files (x86)\MySQL\MySQL Server 5.1\bin\mysql.exe

mysql> INSERT INTO quiz VALUES
-> ('001','Which keyword is used to inherit a class in
'> Java?','implements','extends','inherit','using','extends');
Query OK, 1 row affected (0.03 sec)

mysql>
mysql> INSERT INTO quiz VALUES
-> ('002','Which method is the entry point of a Java
'> program?','start()','main()','run()','init()','main()');
Query OK, 1 row affected (0.04 sec)

mysql>
mysql> INSERT INTO quiz VALUES
-> ('003','Which package contains Scanner
'> class?','java.io','java.util','java.lang','java.sql','java.util');
Query OK, 1 row affected (0.08 sec)

mysql>
mysql> INSERT INTO quiz VALUES
-> ('004','Which is not a primitive data type?','int','float','String','char','String');
Query OK, 1 row affected (0.08 sec)

mysql>
mysql> INSERT INTO quiz VALUES
->
-> ('005','Which exception occurs when a number is divided by
'> zero?','IOException','NullPointerException','ArithmeticException','ClassNotFoundException','
'> ArithmeticException');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM qadb;
ERROR 1146 (42S02): Table 'qadb.qadb' doesn't exist
mysql> SELECT * FROM quiz;
+-----+-----+-----+-----+-----+-----+
| qno | question | op1 | op2 | op3 | op4 | ans |
+-----+-----+-----+-----+-----+-----+
| 001 | Which keyword is used to inherit a class in
Java? | implements | extends | inherit | using | extends |
| 002 | Which method is the entry point of a Java
program? | start() | main() | run() | init() | main() |
| 003 | Which package contains Scanner
class? | java.io | java.util | java.lang | java.sql | java.util |
| 004 | Which is not a primitive data type? | int | float | String | char | String |
| 005 | Which exception occurs when a number is divided by
zero? | IOException | NullPointerException | ArithmeticException | ClassNotFoundException |
ArithmeticException |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

```

AIM(C): Create simple Servlet application to demonstrate Non-Blocking Read Operation.

Code:

```

<!DOCTYPE html>
<html>
<head>
    <title>Non-Blocking Practical</title>
</head>
<body>
    <form action="NBServlet" method="post">
        <h3>Please Click Start Button to Start Non-Blocking Read!!!</h3>
        <input type="submit" value="Start NB IO Read">
    </form>
    <div>TO DO write content</div>
</body>
</html>

```

SERVLET:MyReadListner

```

package EJPracts;
import java.io.IOException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.AsyncContext;

```

```

import javax.servlet.ReadListener;
import javax.servlet.ServletInputStream;
public class MyReadListener implements ReadListener {
    private ServletInputStream iStream = null;
    private AsyncContext cnt = null;
    public MyReadListener(ServletInputStream in, AsyncContext ac) {
        this.iStream = in;
        this.cnt = ac;
    }
    @Override
    public void onDataAvailable() {
        try {
            StringBuilder sb = new StringBuilder();
            int len;
            byte[] b = new byte[1024];
            while (iStream.isReady() && (len = iStream.read(b)) != -1) {
                String mydata = new String(b, 0, len);
                sb.append(mydata);
            }
        } catch (IOException ex) {
            Logger.getLogger(MyReadListener.class.getName()).log(Level.SEVERE, null, ex);
        }
    }
    @Override
    public void onAllDataRead() {
        System.out.println("Called onAllDataRead()");
        cnt.complete();
    }
    @Override
    public void onError(Throwable t) {
        Logger.getLogger(MyReadListener.class.getName()).log(Level.SEVERE, null, t);
        cnt.complete();
    }
}

```

SERVLET:NBServ

```

import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.IOException;
import java.io.InputStream;
import java.io.InputStreamReader;
import java.io.OutputStreamWriter;
import java.io.PrintWriter;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.ServletContext;
import javax.servlet.ServletException;

```

```

import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
@WebServlet(urlPatterns = {"/NBServ"})
public class NBServ extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        String filepath = "/WEB-INF/Introduction.txt";
        ServletContext cnt = getServletContext();
        InputStream iStream = cnt.getResourceAsStream(filepath);
        try (PrintWriter out = response.getWriter()) {
            String path = "http://" + request.getServerName() + ":"
                + request.getServerPort()
                + request.getContextPath()
                + "/ReadNBServ";
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Non-Blocking I/O Servlet</title>");
            out.println("</head>");
            out.println("<body>");
            out.println("<h1>Reading Introduction from file</h1>");
            out.flush();
            URL url = new URL(path);
            HttpURLConnection mycon = (HttpURLConnection) url.openConnection();
            mycon.setChunkedStreamingMode(2);
            mycon.setDoOutput(true);
            mycon.connect();
            if (iStream != null) {
                InputStreamReader isr = new InputStreamReader(iStream);
                BufferedReader br = new BufferedReader(isr);
                String readText;
                try (BufferedWriter bw = new BufferedWriter(
                    new OutputStreamWriter(mycon.getOutputStream()))) {
                    out.println("<div style='width=100%;height:450px;overflow:scroll;'>");
                    while ((readText = br.readLine()) != null) {
                        out.println("<div style='background-color:khaki;width=100%;'>");
                        out.println(readText);
                        out.println("</div><br/>");
                        out.flush();
                        bw.write(readText);
                        Thread.sleep(1000);
                        out.flush();
                    }
                    out.println("</div>");
                    bw.flush();
                }
            }
        }
    }
}

```

```

        }
        out.println("</body>");
        out.println("</html>");
    } catch (InterruptedException | IOException ex) {
        Logger.getLogger(NBServ.class.getName()).log(Level.SEVERE, null, ex);
    }
}

@Override
protected void doGet(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}

@Override
protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}

@Override
public String getServletInfo() {
    return "Short description";
}
}

```

SERVLET:ReadNBServ

```

import EJPracts.MyReadListener;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.AsyncContext;
import javax.servlet.ServletException;
import javax.servlet.ServletInputStream;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet(
    name = "ReadNBServ",
    urlPatterns = {"/ReadNBServ"},
    asyncSupported = true
)

public class ReadNBServ extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            out.println("<!DOCTYPE html>");
            out.println("<html>");

```

```

        out.println("<head>");
        out.println("<title>Servlet ReadNBServ</title>");
        out.println("</head>");
        out.println("<body>");
        AsyncContext cnt = request.startAsync();
        ServletInputStream iStream = request.getInputStream();
        iStream.setReadListener(new MyReadListener(iStream, cnt));
        out.println("</body>");
        out.println("</html>");
    }
}
@Override
protected void doGet(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
@Override
protected void doPost(HttpServletRequest request,
    HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
@Override
public String getServletInfo() {
    return "Short description";
}
}

```

Introduction.txt

Name: Akash Negi

Roll No :T028

Class : TYIT

Output:

Please Click Start Button to Start Non-Blocking Read!!!

Start NB IO Read

TO DO write content

Reading Introduction from file

Name: Sachin Ramesh Mehta

Roll no: T018

Class : TYIT